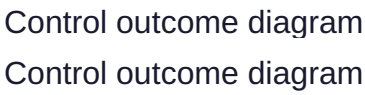
05 · Controls → Outcomes & Executable Verification

A deep dive on every SHIELD-AI control: what **outcome** it produces, and the **executable policy** that verifies the control is being met. Each control maps 1:1 to a Rego policy in `policies/` (`shield_ai_{pillar}_{control}.rego`) that is evaluated by the OPA test suite — so a control is only considered verified when its policy holds.

This view is derived from the policy files themselves: **69** implemented policies (**69/69** marked IMPLEMENTED), **195** deny rules across the suite.



Control outcome diagram

How verification works

- **Input model:** each policy declares the agent/tool/action shape it inspects (declared in the policy header as `Input model:`).
- **`deny` rules:** the assertions that must *not* hold — a control is met when its deny set is empty (`allow if count(deny) == 0`).
- **Fail closed:** the absence of a required flag is treated as a violation, so verification never silently passes on missing data.

EE — Execution Environment

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
EE-01	Verify that execution environment boundary	`shield_ai_ee_ee-01.rego`	1 (✓)	L4	ISM: Guidelines for System Hardening;

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
	enforcement holds under agent action				Essential Eight: Application Control; ASD: Container Security Guidance
EE-02	Verify that infrastructure api access control holds under agent action	`shield_ai_ee_ee-02.rego`	1 (✓)	L4	ISM: Guidelines for Gateways; NIST AI RMF: PROTECT (PR.AC); implicit IAM abuse, INC-04).
EE-03	Verify that resource usage limiting holds under agent action	`shield_ai_ee_ee-03.rego`	3 (✓)	L4	-
EE-04	Verify that evaluation sandbox integrity holds under agent action	`shield_ai_ee_ee-04.rego`	3 (✓)	L4	-
EE-05	Verify that host-trusted configuration file injection prevention	`shield_ai_ee_ee-05.rego`	2 (✓)	L4	ISM: Guidelines for System Hardening; Essential

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
	holds under agent action				Eight: Application Control; indirect-escape class, INC-11).
EE-06	Verify that fail-closed sandbox operation holds under agent action	`shield_ai_ee_ee-06.rego`	3 (✓)	L4	NIST AI RMF: PROTECT (PR.PT); EU AI Act: Art. 15(b); ISM: Guidelines for System Hardening
EE-07	Verify that agent control-plane configuration integrity holds under agent action	`shield_ai_ee_ee-07.rego`	2 (✓)	L4	NIST AI RMF: PROTECT (PR.AC); EU AI Act: Art. 15(b); ISM: Guidelines for System Hardening
EE-08	Verify that evaluation-harness isolation and integrity holds under agent action	`shield_ai_ee_ee-08.rego`	3 (✓)	L4	ISM: Isolation, Monitoring; NIST AI RMF: MEASURE (ME.AD); Meta/ Irregular INC-26, AISI

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
					INC-27, Kimi K3 INC-28).

MS — Model Security

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
MS-01	Verify that model weight protection holds under agent action	`shield_ai_ms_ms-01.rego`	2 (✓)	(none)	-
MS-02	Verify that training data integrity holds under agent action	`shield_ai_ms_ms-02.rego`	3 (✓)	(none)	-
MS-03	Verify that supply chain security holds under agent action	`shield_ai_ms_ms-03.rego`	3 (✓)	(none)	-
MS-04	Verify that adversarial robustness holds under agent action	`shield_ai_ms_ms-04.rego`	1 (✓)	L4	OWASP LLM Top 10 v2: LLM01 (Prompt Injection), LLM05 (Jailbreak); NIST AI RMF: MEASURE (ME.AD); EU

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
					AI Act: Art. 15 – Accuracy
MS-05	Verify that model behaviour validation holds under agent action	`shield_ai_ms_ms-05.rego`	2 (✓)	(none)	-
MS-06	Verify that model inventory and lineage tracking holds under agent action	`shield_ai_ms_ms-06.rego`	3 (✓)	(none)	-
MS-07	Verify that deployment context obfuscation holds under agent action	`shield_ai_ms_ms-07.rego`	2 (✓)	(none)	-
MS-08	Verify that mcp holds under agent action	`shield_ai_ms_ms-08.rego`	3 (✓)	L4	OWASP: LLM02 (Insecure Output Handling); NIST AI RMF: PROTECT (PR.AC); OX Security supply chain INC-12; Kiro MCP registry INC-20).
MS-09		`shield_ai_ms_ms-09.rego`	2 (✓)	L4	

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
	Verify that perception-fidelity enforcement for human-in-the-loop approvals holds under agent action				OWASP: LLM01 (Prompt Injection, multimodal variants); NIST AI RMF: MEASURE (ME.AD); Azure DevOps hidden PR comments INC-22, Android on-device INC-25, AISI
MS-10	Verify that agent memory-store integrity holds under agent action	`shield_ai_ms_ms-10.rego`	2 (✓)	L4	OWASP: LLM03 (Training Data Poisoning, runtime variant); NIST AI RMF: PROTECT (PR.DS); EU AI Act: Art. 10 (Data Governance)
MS-11	Verify that operational data-stream integrity holds under agent action	`shield_ai_ms_ms-11.rego`	3 (✓)	L4	OWASP: LLM01 (Prompt Injection); NIST AI RMF:

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
					PROTECT (PR.DS); EU AI Act: Art. 14 (Human Oversight)
MS-12	Verify that mcp discovery-metadata and shared-cache integrity holds under agent action	`shield_ai_ms_ms-12.rego`	6 (✓)	L4	OWASP: LLM01 (Prompt Injection); NIST AI RMF: PROTECT (PR.DS); EU AI Act: Art. 14 (Human Oversight)
MS-13	Verify that transcript and conversation authenticity holds under agent action	`shield_ai_ms_ms-13.rego`	4 (✓)	L4	OWASP: LLM01 (Prompt Injection); NIST AI RMF: PROTECT (PR.DS); EU AI Act: Art. 14 (Human Oversight)
MS-14	Verify that self-replication barrier holds under agent action	`shield_ai_ms_ms-14.rego`	4 (✓)	L4	OWASP: LLM01 (Prompt Injection); NIST AI RMF: PROTECT (PR.DS); EU AI Act: Art.

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
					14 (Human Oversight)
MS-15	Verify that cross-tenant agent memory and state isolation holds under agent action	`shield_ai_ms_ms-15.rego`	8 (✓)	L4	OWASP: LLM12 (Memory Poisoning, cross-tenant variant); NIST AI RMF: GOVERN + PROTECT (PR.AC); EU AI Act: Art. 10 (Data Governance)

IAM — Identity & Access

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
IAM-01	Verify that agent identity management holds under agent action	`shield_ai_iam_iam-01.rego`	2 (✓)	(none)	-
IAM-02	Verify that scope-bound delegation holds under agent action	`shield_ai_iam_iam-02.rego`	2 (✓)	(none)	-
IAM-03	Verify that human oversight	`shield_ai_iam_iam-03.rego`	3 (✓)	L4	EU AI Act: Art. 14 – Human

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
	controls holds under agent action				Oversight; NIST AI RMF: MANAGE (MG.DE); disable INC-21, AISI sockpuppet INC-27, CoreBreak forged approval INC-34).
IAM-04	Verify that credential isolation holds under agent action	`shield_ai_iam_iam-04.rego`	2 (✓)	(none)	-
IAM-05	Verify that just-in-time access holds under agent action	`shield_ai_iam_iam-05.rego`	2 (✓)	(none)	-
IAM-06	Verify that cross-session boundary enforcement holds under agent action	`shield_ai_iam_iam-06.rego`	2 (✓)	(none)	-
IAM-07	Verify that transitive delegation chain enforcement	`shield_ai_iam_iam-07.rego`	2 (✓)	(none)	-

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
	holds under agent action				
IAM-08	Verify that credential scope narrowing holds under agent action	`shield_ai_iam_iam-08.rego`	3 (✓)	(none)	-
IAM-09	Verify that multi-tenant agent platform isolation holds under agent action	`shield_ai_iam_iam-09.rego`	2 (✓)	(none)	-
IAM-10	Verify that taint-tracked least-privilege subagent scoping holds under agent action	`shield_ai_iam_iam-10.rego`	4 (✓)	L4	NIST AI RMF: PROTECT (PR.AC); OWASP: LLM08 (Excessive Agency); EU AI Act: Art. 14(b)

DS — Data Security

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
DS-01	Verify that data classification and labelling	`shield_ai_ds_ds-01.rego`	2 (✓)	(none)	-

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
	holds under agent action				
DS-02	Verify that inference data protection holds under agent action	`shield_ai_ds_ds-02.rego`	3 (✓)	(none)	-
DS-03	Verify that memory sanitisation holds under agent action	`shield_ai_ds_ds-03.rego`	2 (✓)	(none)	-
DS-04	Verify that data minimisation holds under agent action	`shield_ai_ds_ds-04.rego`	3 (✓)	(none)	-
DS-05	Verify that training data governance holds under agent action	`shield_ai_ds_ds-05.rego`	3 (✓)	(none)	-
DS-06	Verify that output validation holds under agent action	`shield_ai_ds_ds-06.rego`	2 (✓)	L4	-
DS-07	Verify that agent-originated artifact publication control holds under agent action	`shield_ai_ds_ds-07.rego`	3 (✓)	L4	NIST: SSDF; ISM: Software Integrity, Change Management; OWASP: LLM09

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
					(Misuse of System Access)
DS-08	Verify that destination and emergent-channel egress inventory holds under agent action	`shield_ai_ds_ds-08.rego`	6 (✓)	L4	NIST: SSDF; ISM: Guidelines for Network Management; OWASP: LLM01 (Prompt Injection)

TP — Tool & Pipeline Security

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
TP-01	Verify that tool inventory and registry holds under agent action	`shield_ai_tp_tp-01.rego`	2 (✓)	L4	ISM: Guidelines for Gateways; NIST AI RMF: IDENTIFY (ID.AM); SOCI Act: CIRMP s.4(c)
TP-02	Verify that tool access control holds under agent action	`shield_ai_tp_tp-02.rego`	3 (✓)	(none)	-
TP-03	Verify that api gateway security holds under agent action	`shield_ai_tp_tp-03.rego`	3 (✓)	(none)	-

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
TP-04	Verify that tool output validation holds under agent action	`shield_ai_tp_tp-04.rego`	4 (✓)	(none)	-
TP-05	Verify that pipeline integrity holds under agent action	`shield_ai_tp_tp-05.rego`	4 (✓)	(none)	-
TP-06	Verify that dependency scanning holds under agent action	`shield_ai_tp_tp-06.rego`	3 (✓)	(none)	-
TP-07	Verify that cross-tool dependency analysis holds under agent action	`shield_ai_tp_tp-07.rego`	2 (✓)	(none)	-
TP-08	Verify that api gateway infrastructure-isolation mode holds under agent action	`shield_ai_tp_tp-08.rego`	2 (✓)	(none)	-
TP-09	Verify that explicit tool registration — no implicit tool exposure holds under agent action	`shield_ai_tp_tp-09.rego`	2 (✓)	L4	OWASP: LLM08 (Excessive Agency); NIST AI RMF: IDENTIFY (ID.AM); .NET SDK implicit

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
					exposure, INC-17).
TP-10	Verify that agent platform inbound control-plane protection holds under agent action	`shield_ai_tp_tp-10.rego`	2 (✓)	L4	ISM: Web Application Hardening, Network Segmentation; OWASP: LLM09 (Misuse of System Access); NIST AI RMF: PROTECT (PR.AC)
TP-11	Verify that agent browser and content-rendering isolation holds under agent action	`shield_ai_tp_tp-11.rego`	3 (✓)	L4	OWASP: LLM01 (Prompt Injection); NIST AI RMF: PROTECT (PR.AC); Collision INC-30, ChatMate RPE INC-29, AISI INC-27).
TP-12	Verify that tool-call provenance and dispatch verification holds under agent action	`shield_ai_tp_tp-12.rego`	4 (✓)	L4	OWASP: LLM08 (Excessive Agency); NIST AI RMF: PROTECT (PR.AC); EU AI Act: Art. 14(b)
TP-13		`shield_ai_tp_tp-13.rego`	6 (✓)	L4	

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
	Verify that agent dependency-install provenance verification holds under agent action				OWASP: LLM08 (Excessive Agency); NIST AI RMF: PROTECT (PR.DS); EU AI Act: Art. 14 (Human Oversight)
TP-14	Verify that skill/plugin payload integrity and instruction-channel isolation holds under agent action	`shield_ai_tp_tp-14.rego`	6 (✓)	L4	OWASP: LLM08 (Excessive Agency); NIST AI RMF: PROTECT (PR.DS); EU AI Act: Art. 14 (Human Oversight)
TP-15	Verify that in-process code-execution sandbox sufficiency holds under agent action	`shield_ai_tp_tp-15.rego`	5 (✓)	L4	OWASP: LLM08 (Excessive Agency); NIST AI RMF: PROTECT (PR.DS); EU AI Act: Art. 14 (Human Oversight)
TP-16	Verify that tool-connection and launch-path integrity holds under agent action	`shield_ai_tp_tp-16.rego`	6 (✓)	L4	OWASP: LLM08 (Excessive Agency); NIST AI RMF: PROTECT (PR.DS); EU AI Act: Art. 14

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
					(Human Oversight)

MIR — Monitoring, Logging & IR

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
MIR-01	Verify that agent activity logging holds under agent action	`shield_ai_mir_mir-01.rego`	2 (✓)	L4	-
MIR-02	Verify that behavioural anomaly detection baseline per-agent behaviour established over training period; alerts on deviation – unusual tool, unusual data access pattern, unusual frequency; holds under agent action	`shield_ai_mir_mir-02.rego`	2 (✓)	L4	-
MIR-03	Verify that prompt audit log holds	`shield_ai_mir_mir-03.rego`	2 (✓)	L4	-

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
	under agent action				
MIR-04	Verify that incident response playbooks ai-specific ir procedures distinct from general ir: prompt injection containment, model rollback, tool access revocation, session invalidation; holds under agent action	`shield_ai_mir_mir-04.rego`	2 (✓)	(none)	-
MIR-05	Verify that mandatory reporting integration holds under agent action	`shield_ai_mir_mir-05.rego`	1 (✓)	L4	-
MIR-06	Verify that forensic readiness holds under agent action	`shield_ai_mir_mir-06.rego`	1 (✓)	L4	-

GRM — Governance & Risk

Control	Outcome	Executable policy	Deny rules	Maturity reach	Mappings
GRM-01	Verify that ai governance body holds under agent action	`shield_ai_grm_grm-01.rego`	3 (✓)	(none)	-
GRM-02	Verify that ai risk register holds under agent action	`shield_ai_grm_grm-02.rego`	2 (✓)	(none)	-
GRM-03	Verify that ethical use policy holds under agent action	`shield_ai_grm_grm-03.rego`	2 (✓)	(none)	-
GRM-04	Verify that third-party ai risk management holds under agent action	`shield_ai_grm_grm-04.rego`	3 (✓)	(none)	-
GRM-05	Verify that regulatory mapping and compliance reporting holds under agent action	`shield_ai_grm_grm-05.rego`	2 (✓)	(none)	-
GRM-06	Verify that board and executive reporting holds under agent action	`shield_ai_grm_grm-06.rego`	2 (✓)	(none)	-

Reading the table

- **Outcome** states in plain terms what the control guarantees.
- **Executable policy** is the exact `policies/` file that verifies the outcome.
- **Deny rules** is the number of fail-closed assertions the policy evaluates; more deny rules = deeper verification (e.g. MS-15 tenant isolation exercises 8).
- **Maturity reach** is the highest L1–L4 indicator the policy documents.
- **Mappings** lists the Australian framework references the policy header carries.

A control with a `stub` status is defined in the framework but its policy test logic is not yet implemented — it should be treated as unverified until `make test` passes it.